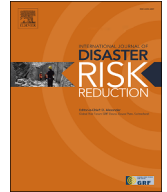




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The current state of early warning system in South Asia: A case study of Afghanistan

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ABSTRACT

This study focused on reviewing flood early warning systems in South Asian countries, using qualitative methods including review of journal articles, case and comparative studies, international frameworks in reference to early warning, and Afghanistan Disaster Management law, policies, and procedures. Employing SWOT analysis, the study identified strengths and weaknesses in Afghanistan's flood early warning system. An analytical comparison matrix was applied to evaluate India, Pakistan, Bangladesh, and Afghanistan's flood hazard early warning systems. Results reveal Afghanistan's end-to-end early warning system predicts flood hazards with a short lead time. In addition, Afghanistan's early warning system weaknesses include data collection and data gaps, while established frameworks and meteorological networks stand as key strengths. The comparison indicates moderate scores i.e. India (Score: 3.695), Bangladesh (Score: 3.317), and Pakistan (Score: 3.15), excluding Afghanistan (Score: 2.378049) highlights that Afghanistan has a relatively weak flood early warning system in South Asia. To improve its EWS, Afghanistan's Disaster Management departments are recommended to get improvements in data integration and dissemination, lead time prediction, capacity building technology advancement, community engagement and ultimately reducing vulnerability.

1. Introduction

Early warning (EW) is a key element of Disaster Risk Reduction (DRR) [1,2]. It consists of risk knowledge, monitoring and prediction, dissemination, and response capability [3]. EW plays a crucial role in disaster preparedness and response activities, saving lives, reducing damage, and enhancing community resilience by sharing timely information with vulnerable communities [4,5].

South Asian countries are exposed to various disasters including earthquakes, landslides, droughts, floods, tsunamis, and cyclones [6]. The frequency and intensity of such disasters are increasing annually [7]. Urban cities in developing countries in South Asia are vulnerable to flood (S. M. H [8]. Economically, India, Pakistan, and Bangladesh suffer major flood losses owing to vulnerability. Poverty and unpreparedness increase susceptibility to natural hazards [9]. India's vulnerability to flash floods requires focused land use planning and engineering for targeted flood protection [10]. Pakistan's combination of mountains and flatlands makes it susceptible

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ble to flooding owing to river overflow and heavy rainfall (S. M. H [8]. Low-lying areas in Bangladesh are vulnerable to flood hazards and have various effects on agriculture and livelihoods [11]. Nepal faces severe flood risks owing to inadequate land-use planning and geological composition [12]. Bhutan faces diverse hazards, such as landslides, floods, fires, windstorms, hailstorms, and earthquakes, having various potential impacts [13]. Floods in Afghanistan directly and indirectly cause loss of livelihood, reduced income, and decreased agricultural production [14]. Insufficient early warnings and low levels of trust are the key contributors to flood damages in Afghanistan [15,16]. Afghanistan, a South Asian country [17], has an arid to semi-arid climate [18], and various hazards such as landslides, earthquakes, avalanches, droughts, and floods [19]. Among them, flood is the most prominent disaster, triggered by heavy rainfall, deforestation, unplanned development, and rapid snow melting [20].

Effective early warnings can mitigate vulnerability [21]. Indonesia has 261 warning receivers and 165 seismometers that are installed for earthquake and tsunami alerts. These systems deliver text and map warnings to the community through social networks [22]. Japan's early warning system includes three main components: observation; forecasting; and products encompassing weekly and seasonal forecasts, warnings, and response operations [23]. India's Central Water Commission (CWC) uses rainfall data for flood forecasting, blending hydrological and hydrodynamic approaches for real-time river predictions. This integrates various rainfall data, including precipitation data, to produce a 5-day forecast (S. R [24]. India's flood forecasting network encompasses 62 major rivers and over 175 stations, providing flood forecasts for all flood-prone states. Data collection employs VHF/HF wireless communication and microcomputers at forecasting centers. Annually, the CWC issues 6000 flood forecasts with over 90 % consistent accuracy, disseminating warnings through various social media platforms (O [25]. To ensure effective flood warnings, the transmission networks to record flood-related data need to be strengthened. Bangladesh's 1-10-day flood-forecasting model predicts hazards and uses local channels, including teachers and religious leaders, as early warning messengers [26]. Pakistan's HEC-HMS model predicts floods by simulating the movement of water during rainfall events. It improves early flood warnings through distributed modeling and continuous simulation [27].

Floods emerged as a significant threat to humans and properties in China between 1990 and 2005, causing approximately 1000 deaths per year. Since 2006, the development of large-scale Flash Flood Warning (FFW) systems has significantly reduced the damage caused by flash floods in China (Zhao et al., 2022). Research has shown that the implementation of flood warning systems effectively reduces the impact of floods in China (Salgado and Nájera, 2022). Real-time monitoring and warning thresholds play a critical role in flood forecasting and timely warnings [28]. Thailand has developed a real-time water monitoring system for flood management, enabling wireless communication between authorities and communities, showcasing the feasibility of current sensor technology [29]. Meteorological and civil protection agencies commonly use rainfall thresholds to issue flood alerts [30]. Iran utilizes the Fuzzy-TOPSIS model for flood warnings based on ensemble forecasts, aiming for a more reliable warning system [31]. China has advanced its flash flood early warning system with real-time monitoring and forecasting to enhance disaster preparedness (Liu et al., 2018). In Northern Henan Province, China, integrating GRACE-FO satellite data and precipitation data has improved the understanding and management of flood events [32]. China's DRIVE-Urban model, a river basin-urban hydrological model, accurately predicts urban flood patterns and aids in flood management and urban planning decisions [33].

The flood end-to-end early warning system in Afghanistan, supported by the World Meteorological Organization and USAID, encompasses data collection, diagnostic tools, warnings, dissemination, and response. Challenges include limited risk awareness in communities, adequate monitoring, and warning dissemination owing to inadequate technological infrastructure, communication, and financial constraints. Therefore, the main aim of this article is to review the flood early warning systems in South Asian countries, with a focus on Afghanistan. The objective is to assess the current status of flood early warnings, identify challenges specific to Afghanistan, and offer recommendations for improvement.

2. Materials and methods

2.1. Study area

This research focuses on reviewing the current state of the Early Warning System (EWS) in South Asia using a specific case study of Afghanistan. India, Pakistan, Bangladesh, and Afghanistan were chosen from South Asia because of their high vulnerability to flooding [34]; S. A. A [35]. India and Pakistan are developing countries [36], whereas Bangladesh and Afghanistan are the least developed [37]. Chosen for their geographical diversity and susceptibility to floods, the flood early warning system in India, Pakistan, Bangladesh, and Afghanistan were reviewed. This review, tailored to their unique climatic conditions and vulnerability to flood security challenges, aims to bolster flood resilience and risk-reduction strategies across South Asia (Fig. 1). The results of this study offer vital insights for policymakers, humanitarian agencies, and stakeholders engaged in disaster management and sustainable development to improved early flood warning efforts.

2.2. Methods

This study conducted a comprehensive review of journal articles, reports, case studies, and official documents through a comparative analysis within Afghanistan's disaster-management domain (see Table 3). The relevant existing literature was identified according to the objectives of this study through a critical review of several studies conducted until 2023. The literature review was performed using online scientific databases such as Springer, Elsevier, and Science Direct through Google Scholar, where the keywords Early Warning System, Flood, South Asia, Afghanistan, and Flood Risk Management were searched. In addition, more than 60 papers and reports have been systematically reviewed to extract the necessary information, with a particular focus on flood early warning in India, Bangladesh, Pakistan, and Afghanistan. Employ a compression matrix for the selected counties' flood early warning systems to identify four fundamental early flood warning components (Risk Knowledge, monitoring, prediction, dissemination, and response ca-

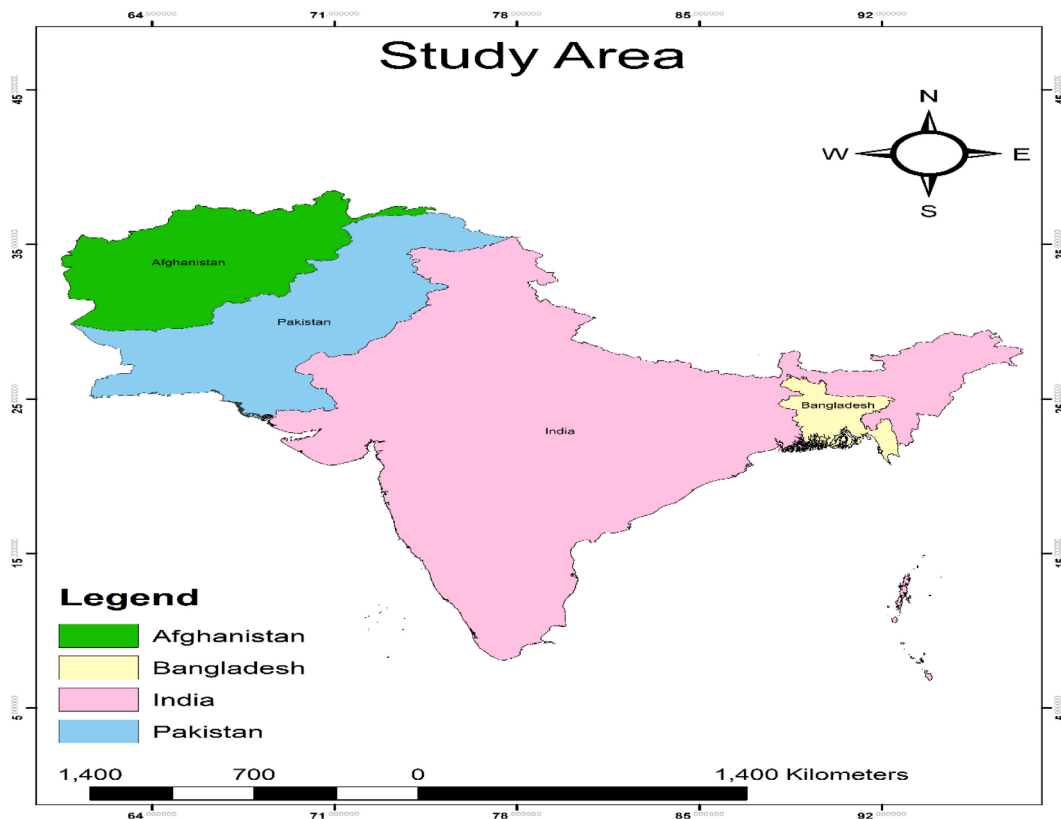


Fig. 1. Study area.

capacity) sourced from literature reviews (Kunguma, 2023). Within each early warning element, 15 sub-indicators were derived, comprising three sub-indicators per main element, gathered through an extensive literature analysis and expert consultations. Seven Disaster Management experts, five affiliated with Kabul University, one from the Afghanistan Metrological Department, and one representing the Afghanistan National Disaster Management Authority (ANDMA), were appointed based on their expertise and knowledge referenced to flood early warning. The experts were tasked with weight and scale considering the range of rank for the selected country's early warning system based on the collected indicators, supportive information checklists, website content, availability of each country's flood early warning data, and based on their technical expertise. In the second stage, to compare flood early warning systems, experts' opinions were selected to set weights (the relative importance of each criterion from 1 was low to 10 was high) by measuring the mean of the experts' opinions to set weights for each indicator. The rating was established ($\text{ratio} = \text{weight of each criterion} / \text{sum} = \text{ratio} * \text{scale}$), and then scales (positive = 4, moderate = 3, negative = 2, unavailable = 1) and scores ($\text{ratio} * \text{scale} = \text{score}$) were calculated for the selected countries. The early flood warning systems are listed in (See Table 4). The expert rating process was finalized within a two-month period. Additionally, Strengths, Weaknesses, Opportunities, and Threats (SWOT) analysis was conducted to determine the strengths and weaknesses of Afghanistan's early flood warning system (Table 1). Finally, the study involved an on-site visit to the Afghanistan Meteorological Department to discuss the nation's flood early warning and prediction system (see Fig. 3).

Table 1
Afghanistan early warning SWOT.

- Established EW framework - Comprehensive system - Technological infrastructure - Metrological stations - Public weather services - Capacity building - Technology advancements - Community engagement - Integration of data	S	W	- Resource constraints - Data gaps - Short-lead time
	O	T	- Political instability - Infrastructure vulnerability - Data privacy and security - Climate change

Table 2

Early warning range of rank.

Range of ranks	
1–1.6	Very weak
1.7–2.2	Weak
2.3–2.8	Moderate
2.9–3.4	Good
3.5–5	Very good

Table 3

Early warning comparison/evaluation indicators.

Indicators	No	Sub-Indicators	Explanations	Recourses
Risk Knowledge	1	Data Availability	Data related to hazards vulnerability and exposure.	[38,39]
	2	Hazard Assessment	Identification of high-risk areas and potential impact of hazards	[40]
	3	Community Awareness	Awareness, and skills related to hazards mitigations	[41]
Monitoring and Prediction	1	Technological Infrastructure	Monitoring networks likewise weather stations	[42]
	2	Data Collection and Analysis	Data collection process, and analysis data in real-time to generate accurate predictions and forecasts.	[43]
	3	Lead-time	Predication lead time between detection and occurrence of an event.	[44]
Dissemination	1	Communication Channels	communication channels; such as text messages, radio broadcasts, and social media	[45]
	2	Accessibility	Access to early warning messages and information	[46]
	3	Cultural Sensitivity	Assess the effectiveness of warning messages in reaching diverse linguistic and cultural groups	[47]
Response capability	1	Community Engagement:	Community engagement in response activities, including drills, evacuation plans, and training.	[48]
	2	Government Preparedness	Assess the readiness of government agencies and emergency responders to react promptly and effectively to warnings.	[49]
	3	Resource Allocation	Allocation of resources, such as funding, personnel, and equipment, to ensure a rapid response.	[50]

Table 4

Early warning comparison in South Asian countries.

Early Warning Criteria Indicators	Early warning sub-indicators	Weight	Ratio	India		Bangladesh		Pakistan		Afghanistan	
				Scale	Score	Scale	Score	Scale	Score	Scale	Score
Risk Knowledge	Data Availability	9	0.11	4	0.44	4	0.44	4	0.44	3	0.33
	Hazard Assessment	6	0.07	4	0.29	4	0.29	3	0.22	3	0.22
	Community Awareness	7	0.09	3	0.26	3	0.26	2	0.17	2	0.17
Monitoring and Prediction	Technological Infrastructure	8	0.10	4	0.39	3	0.29	4	0.39	1	0.10
	Data Collection and Analysis	7	0.09	3	0.26	3	0.26	3	0.26	2	0.17
	Lead time	9	0.11	4	0.44	3	0.33	3	0.33	2	0.22
Flood warning Disseminations	Communication Channels	8	0.10	4	0.39	3	0.29	4	0.39	3	0.29
	Accessibility	6	0.07	4	0.29	4	0.29	3	0.22	2	0.15
	Cultural Sensitivity	5	0.07	4	0.29	4	0.29	4	0.29	4	0.29
Response Capability	Community Engagement:	6	0.07	4	0.29	3	0.22	3	0.22	3	0.22
	Government Preparedness	5	0.06	3	0.18	3	0.18	2	0.12	2	0.12
	Resource Allocation	6	0.07	3	0.22	3	0.22	2	0.15	2	0.15
Sum		82	1		3.74		3.37		3.20		2.43

Flood Early Warning System = FEWS, scales: Positive = 4, Moderate = 3, Negative = 2, Unavailable = 1 Ratio = weight of each criteria/sum = ratio*scale.

3. Results

3.1. Afghanistan flood early warning system

The responsible departments of disaster management in Afghanistan started early flood warning activities in 2011 (ANDMA, 2011) Afghanistan National Disaster Management Authority, 2011). Later on, an end-to-end early warning system of the country was developed by the World Meteorological Organization (WMO) and the USAID Technical Assistance in 2018, as shown in Fig. 2 ("Strength. Hydromet Early Warn. Serv. Afghanistan," 2018). It involved six key sections as follow; (1) Data collection which involved quality-controlled, historical, hydro-meteorological, and real observation data (2) Diagnostics and Prognostics tools such as weather analysis, forecasting, numerical weather prediction and flash flood guidance system (3) Satellite images and numerical weather prediction (4) warning (5) Flood hazards warning dissemination via web-portal, SMS and email (6) Response activities. Afghanistan National Disaster Management Authority (ANDMA), and other national and international organizations are responsible to actively participate in response activities [51].

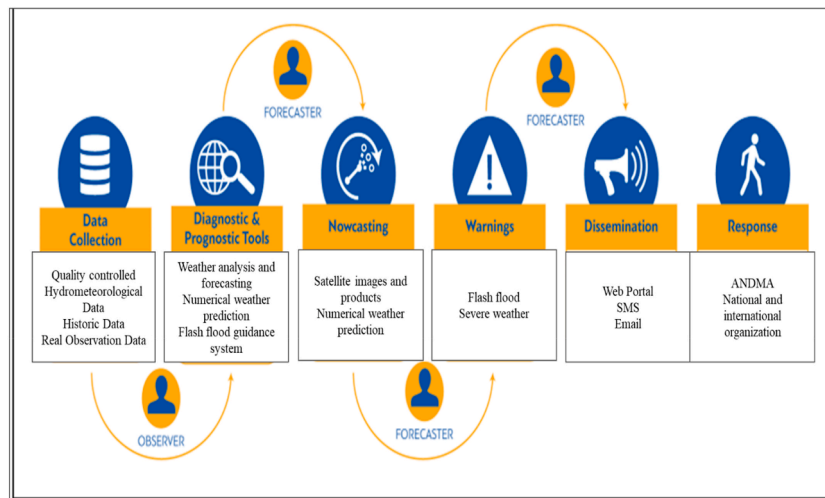


Fig. 2. Afghanistan End to End early warning system.

The Afghanistan Meteorological Department (AMD) is the sole organization responsible for providing public weather services, including precipitation, temperature, and wind parameter-related information. They also play a crucial role in the collection and dissemination of early flood-warning data. Before sharing flood warning information with the public, the AMD collaborated with the Afghanistan National Disaster Management Authority (ANDMA) to ensure a seamless response to flood-related emergencies. Once the AMD officially disseminates the flood early warning, the ANDMA promptly shares the same information with vulnerable communities and simultaneously oversees flood response activities in the country.

Upon early flood warnings from the Afghanistan Meteorological Department (AMD), flood response duties rest with ANDMA, line ministries, and the Disasters Management Higher Commission. Since 2000, the international community has provided financial and technical support to diverse sectors of Afghanistan, including disaster management. This flood early warning system incorporates the Afghanistan Disaster Management Organization along with local and international entities.

The dissemination of flood forecasting data in Afghanistan by the Afghanistan Meteorological Department and ANDMA via web portals, SMS, and email was found to be ineffective due to socio-economic constraints and low community awareness. In Kabul Province, 69 % of the communities were unfamiliar with early flood warnings, and 56 % did not receive any such information. Traditional methods, such as sharing information through education centers and mosques, have been found to be much more effective [52,53]. In Kabul, 76 % of communities expressed willingness to receive information through mosques and social media, emphasizing the community's readiness to use these traditional channels for information dissemination [52]. The ANDMA and other line ministries are responsible for countrywide disaster response efforts, and the coordination of these efforts falls under the purview of the Afghanistan Disaster Management Commission led by the first deputy Prime Minister of the Taliban government. The Afghanistan Disaster Management Law (2019), Afghanistan National Disaster Management Plan (2011), and Disaster Management Guidelines are prominent policy frameworks in this context [54]; 2012). During the Taliban government in Afghanistan, the disaster management and early warning activities were overseen by the Afghanistan Meteorological Department and ANDMA. Issues surrounding international recognition by the current government have hindered their involvement in the development sectors, including disaster management. Despite these challenges, a number of international organizations continue to aid Afghan communities during emergencies. The earthquake in Herat Province in October 2023 received substantial support from UN agencies, international organizations, and Afghanistan's private sector. Collaborative efforts between the international community and Afghan entities highlight the necessity for the Afghan government to enhance collaboration with international bodies to fortify their disaster management, effectively addressing challenges experienced during emergencies.

3.2. Afghanistan meteorological network

The infrastructure of Local Area Network (LAN) of Afghanistan Meteorological Department (AMD) has 10 facilities to exchange data through the World Meteorological Organization (WMO) Global Telecommunication System (GTS) and File Transfer Protocol (Fig. 3). AMD has interconnected server for Pakistan-Afghanistan Flash Flood Guidance System, Numerical Weather Prediction (NWP) centers, and regional meteorological centers [55]. Previously, AMD had lack of access to the WMO GTS for sharing observational reports, similar to surface synoptic observations data with other WMO Members. Through this initiative, AMD possesses essential meteorological data processing and visualization with WMO. This initiative empowered AMD to send and receive GTS data efficiently via Turkish State Meteorological Service Message System. The AMD surface meteorological network includes Five AWOS installations and upgrades to three manual stations. Each AWOS station measured 15 parameters: including pressure, air temperature, precipitation, soil moisture and temperature, wind speed, and direction. These parameters aid flood hazard prediction in Afghanistan.

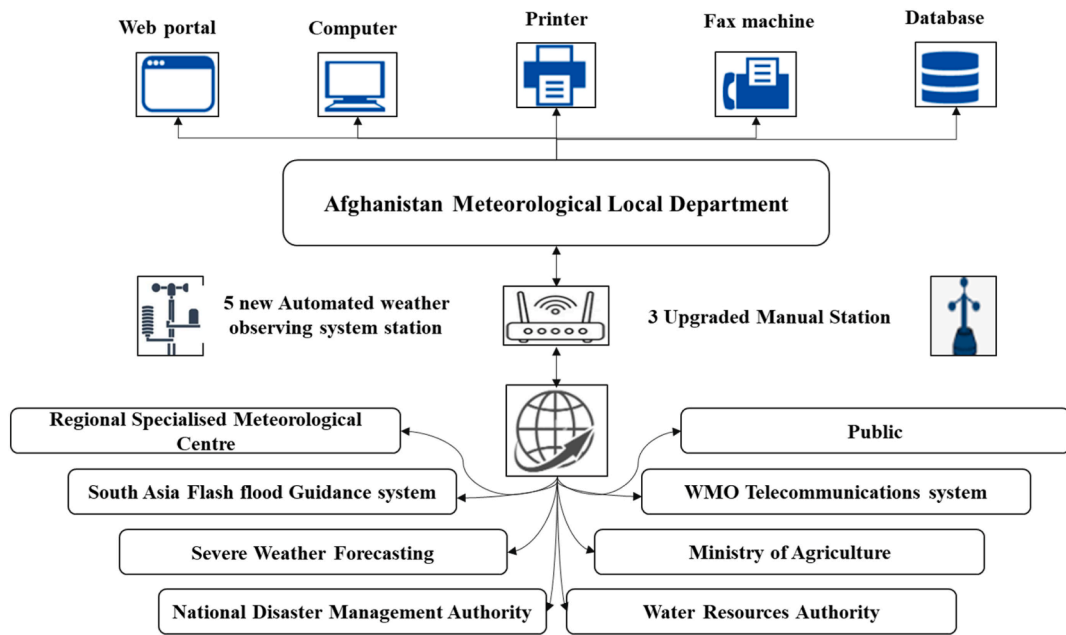


Fig. 3. Afghanistan meteorological departments <https://public.wmo.int/en/projects/Afghanistan-EWS>.

3.3. Public weather service

Since 1990 Afghanistan Meteorological Department (AMD) delivers dependable public weather services to community such as severe weather, flash flood predictions and early warning. Utilizing METCAP + software in daily operations, forecasts and warnings are formulated and communicated to relevant organizations, including Afghanistan Disaster Management Authority via email, fax, and phone. Additionally, a dedicated web portal offers public access to forecasts, alerts, and current weather data (Fig. 4). The issue of false flood alerts in communities is a major concern (Bartholmes et al., 2009). It is crucial to have a range of flood hazard measures for public safety (Abd-Elaty et al., 2023) including community participation in early flood warnings (Rijal, 2023), raising awareness, and promoting self-preparedness (Xie et al., 2023). The establishment of village civil protection committees is vital for mobilizing stakeholders during flood emergencies (Dewa et al., 2023). Understanding flood warning signs, effective communication, and raising awareness through campaigns play a significant role in community public safety (Toyoda & Tanwattana, 2023), (Shah et al., 2023), (Bera, 2023). Collaboration between community-based flood management groups and governmental and non-governmental institu-

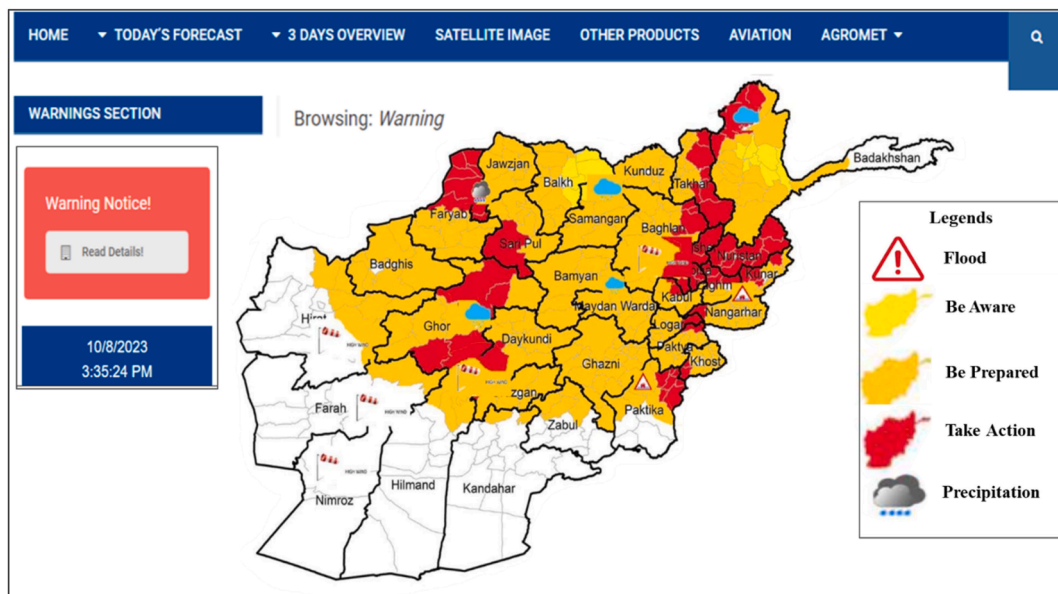


Fig. 4. AMD Early Warning source <http://www.amd.gov.af/category/warning/>.

tions is another crucial aspect that helps the community ensure safety (Li & Lin, 2023), (Jose Moises & Kunguma, 2023) During flood warnings, villagers must adhere to specific guidelines (Bera, 2023). Evacuation centers are a major challenge during flood hazard emergencies in affected communities. According to the Afghanistan Disaster Management Law, the Afghanistan Disasters Management Authority can use public infrastructure and buildings such as schools and universities during emergencies. However, the public community is required to use basic public safety measures, such as evacuation routes, emergency services, and access to essential supplies and facilities, to improve their safety in cases of false or missed flood alerts (Zhong et al., 2013)

3.4. Strength, weakness, opportunity and treats (SWOT) of Afghanistan early warning system

Strength, weakness, opportunities and treat (SWOT) is an effective analysis to uncover strengths and weaknesses of early warning systems [56]. Therefore, this approach was applied to evaluate the strengths and weaknesses of Afghanistan's early warning system. The strengths of the meteorological network include an established early warning framework, a comprehensive approach covering data collection, analysis, and response, and a technologically advanced AMD infrastructure. This infrastructure also facilitates public services such as precise weather predictions. However, limitations in funding, expertise, and data gaps are key weaknesses. Opportunities lie in capacity-building capacity, utilizing technology for accurate weather data, and community participation for effective preparedness and response. The network faces threats from political instability affecting funding and collaboration, infrastructure vulnerabilities, data security concerns, and climate change impacts.

3.5. South Asian countries flood early warning system: a comparison with Afghanistan

The Afghanistan flood early warning system is analytically compared with South Asian countries such as Bangladesh, India, and Pakistan by utilizing an analytical comparison matrix (See Table 4)). For the comparison of flood Early warning systems (EWS) in South Asian countries in, the main elements include risk knowledge, monitoring and predication, dissemination, and response capacity (Table 2) [3,16,57]. For each element of the EWS, three sub-indicators and overall twelve indicators were identified and selected. In the second stage, the experts set the weights for all indicators. The groups of experts rated the indicators from 1 to 10 where 1 was low and 10 was high. The rating was established (ratio = weight of each criterion divided by sum). Then, the scales (positive = 4, moderate = 3, not access = 2, negative = 1) and scores (score = ratio $\times$ scale) for selected countries were calculated (Table 2). The range of the early warning for the analytical comparison was ranked as very weak, weak, moderate, good, and very good [58].

Table (4) indicates the comparison of Early Warning Systems (EWS) among South Asian countries using four key elements as criteria and 12-sub-indicators. The analytical matrix shows the effectiveness of each country's EWS based on a set of criteria, sub-indicators, weights, and scales. Criteria and Sub-Indicators represent key aspects of EWS evaluation, and the weight shows the importance of the sub-indicator of early warning. The Scale shows a qualitative value to each sub-indicator (positive = 4, moderate = 3, negative = 2, and unavailable = 1) based on its effectiveness. The score for each sub-indicator was calculated by multiplying its weight by its corresponding scale value. The sum of the scores within each criterion provides a cumulative measure of performance for that criterion. The ratio for each criterion was calculated by dividing the weights by the sum of all weights, representing its contribution to the overall assessment. The sum of all criteria scores and the sum of all criteria ratios represent an aggregate measure of EWS effectiveness for each country. Finally, the result shows based on the overall score that India with 3.74 has a Moderate = 3 early warning system, Bangladesh has a 3.37 overall score of Moderate = 3, Pakistan has an overall score of 3.20 has a Moderate = 3 and Afghanistan has a total score of 2.43 Negative = 2 early warning system. The Afghanistan early warning comparison score is relatively low compared to that of India, Pakistan, and Bangladesh. The Technological infrastructure, prediction lead time, data analysis, community accessibility, resource allocation, and preparedness indicators had low scores; therefore, there is a need for the Afghan government to improve in these areas.

4. Discussion

Afghanistan's flood early warning system boasts a robust framework and technological infrastructure, enabling communication with vulnerable communities through web portals, SMS, and email. However, studies indicate its ineffectiveness, as most prefer traditional methods such as educational centers, mosques, and local media for warnings. This preference resonates with approaches seen in countries such as Bangladesh. The lack of community awareness regarding floods compounded the system's challenges, alongside resource limitations, data gaps, and short lead times.

Comparatively, India's early flood warning system stands out with a 5-day or longer and high accuracy (O [25,59]. Bangladesh's early warning lead time spans 1-10 days, employing various dissemination channels such as educational centers, mosques, local radio, and community leaders [26,60]. These methods have proven effective, especially in less developed countries, where low awareness and financial constraints exist. In contrast, Afghanistan faces constraints, with limited forecasting lead times (no less than five days) and less effective communication methods such as web portals, SMS, and email. Previous studies in Kabul revealed that the majority of communities did not receive early flood warning messages, and over 50 % were unfamiliar with the system. To optimize response times, a reliable wireless communication system lacking in Afghanistan's flood early warning setup is imperative. Unlike Pakistan, India, Bangladesh, Thailand, and China, which have sufficient meteorological stations and services crucial for successful early flood warnings, Afghanistan faces limitations in hydro meteorological services.

Furthermore, other Asian countries possess adequate post-warning flood response facilities. However, Afghanistan's Disaster Management capacity is limited, with ANDMA being the sole organization responsible for response activities. Key limitations in Afghanistan include low disaster management awareness, political instability, limited technical capacity, financial constraints, and is-

sues arising from the international community's negotiations with the Taliban government since 2021, leading to an exodus of experts.

Overall, Afghanistan lags behind its regional counterparts in risk knowledge, prediction, monitoring, dissemination, and response. Further research is necessary to identify suitable communication methods tailored to the community's needs. Urgent support from flood management departments is imperative to fortify Afghanistan's early warning system and mitigate community impacts.

5. Conclusion

Effective early warning systems play a crucial role in disaster management. Afghanistan's comprehensive end-to-end early warning system compressed data collection, forecasting tools, dissemination of warning, and response capabilities. The system exhibits strengths, such as established infrastructure, while data gaps and resources limitations emerge as notable weaknesses. Moreover, challenges include insecurity, political instability, and threats from climate change. Since India, Pakistan, and Bangladesh are using a matrix, it was evident that Afghanistan's early warning system is comparatively weaker than other South Asian countries. Bangladesh notably boasts a long lead time of 7–10 days for flood prediction, employing traditional methods like schools and mosques for effective flood information dissemination. Conversely, Afghanistan relies on media platforms due to financial constraints, limited literacy, and technology access, rendering these methods less feasible nationwide.

As there underscore the urgency for Afghanistan's Disaster Management authorities to enhance the flood early warning system. Adoption of more effective and accessible communication methods is vital. Further research is imperative to identify optimal dissemination techniques that suit the unique context and needs of the region.

CRedit authorship contribution statement

Mokbul Ahmad Morshed: Resources, Supervision, Writing - review & editing. **Hayatullah Mushwani:** Conceptualization, Writing - original draft. **Kawoon Sahak:** Data curation, Resources, Software, Validation, Visualization. **Mohammad Haroon Hairan:** Data curation, Formal analysis, Investigation, Software, Validation.

Declaration of competing interest

I, [Hayatullah Mushwani], declare that there are no conflicts of interest to disclose regarding the research presented in the manuscript titled "The Current State of Early Warning System in South Asia: A Case Study of Afghanistan."

Data availability

The data that has been used is confidential.

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